



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

API GATEWAY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Architecture

TOTAL: 10 QUESTIONS

**Q1** What is API Gateway and what problem in Architecture does it solve?

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**Q6** How does API Gateway handle reliability and high availability?

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**Q2** What are the core components or concepts in API Gateway?

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**Q7** What observability does API Gateway provide out of the box?

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**Q3** How does API Gateway compare to its main alternatives?

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**Q8** What are common performance bottlenecks in API Gateway?

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**Q4** How do you get started with API Gateway for a new project?

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**Q9** What security best practices apply when running API Gateway?

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**Q5** What configuration options most affect API Gateway's behavior in production?

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**Q10** What mistakes do teams commonly make when adopting API Gateway?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 API Gateway is a tool or platform

Mitigation

API Gateway is a tool or platform that automates or simplifies a specific concern in the Architecture domain, reducing manual effort and operational complexity for engineering teams.

A6 API Gateway provides HA through leader election,

Observability

API Gateway provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 API Gateway has key building blocks that

Primitives

API Gateway has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 API Gateway exposes metrics (Prometheus endpoint or

Automation

API Gateway exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 API Gateway trades off simplicity against feature

Hardening

API Gateway trades off simplicity against feature richness differently than its alternatives. Choose API Gateway when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install API Gateway via its package manager

Gotchas

Install API Gateway via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run API Gateway with the principle of

Assessment

Run API Gateway with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.